Code (DEC)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Code (BIN)	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
U_out [V]	−0,000541	0,373151	0,653508	1,027	1,261	1,635	1,915	2,289	2,499	2,873	3,153	3,527	3,761	4,134	4,415	4,789

1. What is the role of the opamp in the examined R-2R ladder circuit?

The opamp converts the current from the R-2R ladder into an output voltage. It also helps to keep the output stable and makes it easier to measure the voltage.

2. Demonstrate your understanding of the circuit to the lab instructor.

The R-2R ladder converts digital signals into analog voltage. In our circuit, we have 4 switches, and each switch represents one bit. When we change the switches, the output voltage also changes. The first bit (MSB) has the biggest effect on the voltage, while the last bit (LSB) has the smallest effect.

In my simulation, the output voltage increased from almost 0 V for 0000 to 4.789 V for 1111.

3. Compare the circuit to an alternative R-2R ladder configuration.

How do these circuits differ?

Our circuit uses an opamp to convert current into voltage. Another R-2R ladder configuration can give an output voltage directly without using an opamp.

What could be the advantages and disadvantages of either configuration?

The main advantage of our circuit is that the opamp helps to provide a stable output and reduces the effect of the connected load. The disadvantage is that it needs more components and an additional power supply.

The alternative circuit is simpler because it does not need an opamp. However, the output voltage can change depending on the connected load.

Task 2

1. How does the ADC circuit work?

The ADC converts an analog voltage into a digital value. The 8-bit counter counts up, and the DAC converts this number into a voltage. The comparator compares the DAC voltage with the input voltage. When the DAC voltage reaches the input voltage, the counter stops, and the display shows the digital value.

2. What happened when changing the input voltage?

When I changed the input voltage, I noticed that the number on the ADC display also changed. For example, at 2.873 V, the display showed 137, and at 1.089 V, it showed 56. So when the input voltage increased, the digital value also increased.

3. How did I organize the circuit blocks?

I moved the blocks to make the circuit easier to understand. I placed the Analogue Voltage Source first, then the Comparator, 8-bit Counter and ADC Display. I placed the DAC above them because it takes the digital value from the counter and converts it into a voltage that goes back to the comparator.

Task3

Experimental setup

In this task, I used the NI USB-6211 to test the ADC and DAC. I connected AO0 to AI0 and AOGND to AIGND.

In LabVIEW, I generated 500 voltage samples from 0 to 0.001 V using Ramp by Samples. I used RSE mode and selected the input range from -0.2 V to 0.2 V. The generated and measured voltages were displayed on an XY Graph.

Results and observations

The graph showed that the measured voltage increased when the generated voltage increased. However, the signal was not smooth, and I could see some noise and voltage steps.

From the data, I estimated the LSB to be around 0.316 mV. I also observed a negative voltage offset of approximately -0.708 mV at the beginning of the measurement. Because of the noise, it was difficult to estimate the exact gain error.

Improving accuracy

One way to improve the accuracy would be to repeat the measurements and calculate the average voltage for each level. This would help reduce the influence of noise and make the voltage steps easier to identify.

Output voltage after stopping

After stopping the LabVIEW program, I measured approximately 102 µV at AI0, which was connected to AO0. This value was close to 0 V, although some noise was still visible.

Conclusion

In this lab, I learned how DAC and ADC circuits work using Falstad simulations and how to test their performance with LabVIEW and NI USB-6211. I also learned how to generate voltage signals, measure them and save the results for analysis.